Chapter 4. Sampling Distributions

Distributions of statistics, chi-square, t and F, order statistics, multivariate normal

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4.1 Distributions of statistics

A **random sample** $X_1, \dots, X_n$ is i.i.d. from a population pdf f; a **statistic** is a function of the sample not involving unknown parameters, and its distribution is its **sampling distribution**. For instance, for a Bernoulli(p) sample, $n\hat{p} = \sum X_i \sim B(n,p)$, so $E\hat{p} = p$ and $\operatorname{Var}\hat{p} = p(1-p)/n$: $\hat{p}$ concentrates around p.

Discrete case. If $Y = u(\mathbf{X})$, then $f_Y(y) = \sum_{\mathbf{x}:u(\mathbf{x})=y} f_{\mathbf{X}}(\mathbf{x})$. Convolution sums reproduce the closure of binomial and Poisson families under addition.

Continuous case: change of variables.

Theorem 4.1

Let $\mathbf{X}$ have pdf $f_{\mathbf{X}}$ supported on an open set $\mathcal{X} \subseteq \mathbb{R}^n$, and let $u : \mathcal{X} \to \mathcal{Y}$ be one-to-one and onto with differentiable inverse $\mathbf{x} = u^{-1}(\mathbf{y})$ and Jacobian $J(\mathbf{y}) = \det(\partial x_i/\partial y_j) \neq 0$. Then $\mathbf{Y} = u(\mathbf{X})$ has pdf

$$f_{\mathbf{Y}}(\mathbf{y}) = f_{\mathbf{X}}(u^{-1}(\mathbf{y}))\,|J(\mathbf{y})|, \qquad \mathbf{y} \in \mathcal{Y}.$$

If instead $\mathcal{X}$ splits into pieces $\mathcal{X}_1, \dots, \mathcal{X}_m$ on each of which u is one-to-one onto $\mathcal{Y}$, then $f_{\mathbf{Y}}(\mathbf{y}) = \sum_j f_{\mathbf{X}}(u_j^{-1}(\mathbf{y}))|J_j(\mathbf{y})|$.

Proof. For $B \subseteq \mathcal{Y}$, $P(\mathbf{Y} \in B) = P(\mathbf{X} \in u^{-1}(B)) = \int_{u^{-1}(B)} f_{\mathbf{X}}(\mathbf{x})\,d\mathbf{x} = \int_B f_{\mathbf{X}}(u^{-1}(\mathbf{y}))|J(\mathbf{y})|\,d\mathbf{y}$ by the change-of-variables formula for integrals. In the many-to-one case, add the contributions of the pieces. ■

To get the pdf of a single $Y_1 = u_1(\mathbf{X})$, complete it to a one-to-one map $(Y_1, \dots, Y_n)$ by adding convenient coordinates, apply the theorem, then integrate out the extra coordinates.

Example 4.1

$X \sim N(0,1)$ and $Y = X^2$: the map $x \mapsto x^2$ is two-to-one, with branches $x = \pm\sqrt{y}$ and $|J| = \frac{1}{2\sqrt{y}}$. So $f_Y(y) = 2\phi(\sqrt{y})\frac{1}{2\sqrt{y}} = \frac{1}{\sqrt{2\pi}} y^{-1/2} e^{-y/2}$, $y > 0$: the Gamma$(\frac{1}{2}, 2)$ density.

Example 4.2 (Beta and gamma)

Let $X_1 \sim \text{Gamma}(\alpha_1, 1)$ and $X_2 \sim \text{Gamma}(\alpha_2, 1)$ be independent, $Y_1 = X_1 + X_2$ and $Y_2 = X_1/(X_1 + X_2)$. The inverse is $x_1 = y_1 y_2$, $x_2 = y_1(1 - y_2)$ with $|J| = y_1$, so

$$f(y_1, y_2) = \frac{y_1^{\alpha_1+\alpha_2-1} e^{-y_1}}{\Gamma(\alpha_1+\alpha_2)} \cdot \frac{\Gamma(\alpha_1+\alpha_2)}{\Gamma(\alpha_1)\Gamma(\alpha_2)} y_2^{\alpha_1-1}(1-y_2)^{\alpha_2-1}.$$

Hence $Y_1 \sim \text{Gamma}(\alpha_1 + \alpha_2, 1)$ and $Y_2 \sim \text{Beta}(\alpha_1, \alpha_2)$, and they are independent. As a by-product, $B(\alpha_1, \alpha_2) = \int_0^1 y^{\alpha_1-1}(1-y)^{\alpha_2-1} dy = \Gamma(\alpha_1)\Gamma(\alpha_2)/\Gamma(\alpha_1+\alpha_2)$.

Example 4.3

For $X_1, X_2 \overset{\text{i.i.d.}}{\sim} U(0,1)$, the mean $(X_1 + X_2)/2$ has the triangular density $2 - 4|y - \frac{1}{2}|$ on $(0,1)$. (Use $Y = (X_1 + X_2)/2$, $Z = (X_1 - X_2)/2$, $|J| = 2$, and integrate out z.) In general $U(a,b) \overset{d}{=} a + (b-a)U(0,1)$.

Probability integral transform. If F is continuous and $X \sim F$, then $F(X) \sim U(0,1)$. Conversely, if $U \sim U(0,1)$, then $F^{-1}(U) \sim F$ with $F^{-1}(u) = \inf\{x : F(x) \geq u\}$. This is the basis of

simulation.

4.2 Chi-square, t and F distributions

Definition (Chi-square)

$\chi^2(r) = \text{Gamma}(r/2, 2)$. For integer r, $\chi^2(r) \overset{d}{=} Z_1^2 + \cdots + Z_r^2$ with $Z_i \overset{\text{i.i.d.}}{\sim} N(0,1)$ (Example 4.1 and the gamma sum rule).

So $E = r$, $\text{Var} = 2r$, $M(t) = (1-2t)^{-r/2}$, and independent chi-squares add their degrees of freedom. The upper α point $\chi^2_\alpha(r)$ satisfies $P(Y > \chi^2_\alpha(r)) = \alpha$. Since $X \sim \text{Gamma}(\nu, \beta) \iff 2X/\beta \sim \chi^2(2\nu)$, gamma quantiles are $\frac{\beta}{2}\chi^2_\alpha(2\nu)$.

Theorem 4.2 (Normal samples)

Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} N(\mu, \sigma^2)$, $\bar{X} = \frac{1}{n}\sum X_i$, $S^2 = \frac{1}{n-1}\sum (X_i - \bar{X})^2$. Then

(a) $\bar{X} \sim N(\mu, \sigma^2/n)$;

(b) $\bar{X}$ and S^2 are independent;

(c) $(n-1)S^2/\sigma^2 \sim \chi^2(n-1)$.

Proof. Standardize so that $\mu = 0$ and $\sigma = 1$, and choose an orthogonal matrix P whose first row is $(1/\sqrt{n}, \ldots, 1/\sqrt{n})$ and set $\mathbf{Y} = P\mathbf{X}$. Since $\mathbf{X} \sim N_n(\mathbf{0}, I)$, also $\mathbf{Y} \sim N_n(\mathbf{0}, PP') = N_n(\mathbf{0}, I)$ (Section 4.4), so $Y_1, \ldots, Y_n$ are i.i.d. $N(0,1)$. Now $Y_1 = \sqrt{n}\bar{X}$, and $\sum_i Y_i^2 = \sum_i X_i^2$, so

$$\sum_{i \geq 2} Y_i^2 = \sum X_i^2 - n\bar{X}^2 = (n-1)S^2.$$

. Hence $\bar{X}$ depends only on Y_1 and S^2 only on $Y_2, \ldots, Y_n$, which gives (b). Also $(n-1)S^2$ is a sum of $n-1$ squared standard normals, giving (c). (a) holds since $Y_1 \sim N(0,1)$. ■

Definition (Student's t and Snedecor's F)

If $Z \sim N(0,1)$ and $V \sim \chi^2(r)$ are independent, then $T = Z/\sqrt{V/r} \sim t(r)$. If $V_i \sim \chi^2(r_i)$ are independent, then $F = (V_1/r_1)/(V_2/r_2) \sim F(r_1, r_2)$.

Proposition 4.3

$$f_{t(r)}(x) = \frac{\Gamma(\frac{r+1}{2})}{\sqrt{r\pi}\,\Gamma(\frac{r}{2})}\left(1 + \frac{x^2}{r}\right)^{-\frac{r+1}{2}},$$

$$f_{F(r_1,r_2)}(x) = \frac{\Gamma(\frac{r_1+r_2}{2})}{\Gamma(\frac{r_1}{2})\Gamma(\frac{r_2}{2})}\left(\frac{r_1}{r_2}\right)^{r_1/2} \frac{x^{r_1/2-1}}{(1 + \frac{r_1}{r_2}x)^{(r_1+r_2)/2}}.$$

Moreover $T^2 \sim F(1, r)$, $1/F(r_1, r_2) \sim F(r_2, r_1)$, and $t(r) \to N(0,1)$ as $r \to \infty$.

Proof. For t: map $(Z, V) \mapsto (T, V)$ with $z = t\sqrt{v/r}$, $|J| = \sqrt{v/r}$, and integrate v out using the gamma integral. For F: map $(V_1, V_2) \mapsto (F, V_2)$ with $v_1 = r_1 f v_2/r_2$, $|J| = r_1 v_2/r_2$, and integrate v_2 out. $T^2 = Z^2/(V/r)$ with $Z^2 \sim \chi^2(1)$. Finally $V/r \to 1$ (its variance is $2/r \to 0$), so $T \to Z$ in distribution (Chapter 5). ■

Corollary 4.4 (Inference for a normal mean and variance)

For a normal sample, $\sqrt{n}(\bar{X}-\mu)/S \sim t(n-1)$ and $(n-1)S^2/\sigma^2 \sim \chi^2(n-1)$. Hence the $1-\alpha$ confidence intervals

$$\mu \in \bar{X} \pm t_{\alpha/2}(n-1)\frac{S}{\sqrt{n}}, \qquad \sigma^2 \in \Big(\frac{(n-1)S^2}{\chi^2_{\alpha/2}(n-1)}, \frac{(n-1)S^2}{\chi^2_{1-\alpha/2}(n-1)}\Big).$$

For two independent normal samples, $\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F(n_1-1, n_2-1)$; with a common variance, the pooled statistic $\frac{\bar{X}_1-\bar{X}_2-(\mu_1-\mu_2)}{S_p\sqrt{1/n_1+1/n_2}} \sim t(n_1+n_2-2)$, where $S_p^2 = \frac{(n_1-1)S_1^2+(n_2-1)S_2^2}{n_1+n_2-2}$.

Proof. $\sqrt{n}(\bar{X}-\mu)/\sigma \sim N(0,1)$ is independent of $(n-1)S^2/\sigma^2 \sim \chi^2(n-1)$ (Theorem 4.2); their ratio is $t(n-1)$. The intervals invert the probability statements, e.g. $P(\chi^2_{1-\alpha/2} \le (n-1)S^2/\sigma^2 \le \chi^2_{\alpha/2}) = 1-\alpha$. For two samples, the chi-squares are independent and add. ■

More generally, in the one-way model with independent $X_{ij} \sim N(\mu_i, \sigma^2)$, $i = 1,\dots,k$, $j = 1,\dots,n_i$, the group means are independent of the within-group sum of squares $\sum_{i,j}(X_{ij}-\bar{X}_i)^2/\sigma^2 \sim \chi^2(n-k)$. Also $\sum_i n_i(\bar{X}_i - \bar{X} - (\mu_i-\bar{\mu}))^2/\sigma^2 \sim \chi^2(k-1)$ (proved via Theorem 4.9 below).

4.3 Order statistics

Let $X_1,\dots,X_n$ be a sample from a continuous pdf f with cdf F, and $X_{(1)} < \cdots < X_{(n)}$ the order statistics (ties have probability 0).

Theorem 4.5

The joint pdf of $(X_{(1)},\dots,X_{(n)})$ is $n!\, f(y_1)\cdots f(y_n)$ on $y_1 < \cdots < y_n$.

Proof. The map from $\mathbf{x}$ (with distinct coordinates) to its sorted version is $n!$-to-one. On each of the $n!$ regions $\{x_{\pi 1} < \cdots < x_{\pi n}\}$ it is a permutation of coordinates with $|J| = 1$. Apply Theorem 4.1 and add the $n!$ equal terms. ■

Theorem 4.6

$$f_{X_{(r)}}(y) = \frac{n!}{(r-1)!(n-r)!}F(y)^{r-1}f(y)\{1-F(y)\}^{n-r},$$

and for $r < s$,

$$f_{X_{(r)},X_{(s)}}(y,z) = \frac{n!}{(r-1)!(s-r-1)!(n-s)!} \times F(y)^{r-1}f(y)\{F(z)-F(y)\}^{s-r-1}f(z)\{1-F(z)\}^{n-s}, \quad y < z.$$

Proof. $X_{(r)} \in (y, y+dy]$ requires $r-1$ observations below y, one in $(y, y+dy]$ and $n-r$ above: a multinomial event with probability $\frac{n!}{(r-1)!1!(n-r)!}F^{r-1}(f\,dy)(1-F)^{n-r}$. The joint case uses five cells in the same way. (Alternatively integrate Theorem 4.5.) ■

In particular $X_{(n)}$ has cdf F^n and $X_{(1)}$ has cdf $1-(1-F)^n$. Also $P(X_{(r)} \le y) = P(\#\{X_i \le y\} \ge r) = \sum_{k \ge r}\binom{n}{k}F(y)^k(1-F(y))^{n-k}$.

Example 4.4

For a $U(0,1)$ sample, $X_{(r)} \sim \text{Beta}(r, n-r+1)$, so $EX_{(r)} = \frac{r}{n+1}$. Since $F(X_i) \sim U(0,1)$ for any continuous F, $F(X_{(r)}) \sim \text{Beta}(r, n-r+1)$ always, which underlies distribution-free intervals for quantiles. For $\text{Exp}(\beta)$ samples, $X_{(1)} \sim \text{Exp}(\beta/n)$. More generally the spacings $(n-i+1)(X_{(i)} - X_{(i-1)})$ are i.i.d. $\text{Exp}(\beta)$; this follows from Theorem 4.5 with a linear change of variables of unit Jacobian after scaling.

The **sample median** is $X_{((n+1)/2)}$ (n odd) and the **range** is $X_{(n)} - X_{(1)}$; their distributions follow from Theorem 4.6 by a further change of variables.

4.4 The multivariate normal distribution

Definition (Multivariate normal)

$\mathbf{X} \sim N_n(\mu, \Sigma)$ if $\mathbf{X} \overset{d}{=} A\mathbf{Z} + \mu$ for some $n \times m$ matrix A with $AA' = \Sigma$ and $\mathbf{Z} \sim N_m(\mathbf{0}, I)$ (i.i.d. standard normals). Equivalently, its mgf is $M(\mathbf{t}) = \exp(\mu'\mathbf{t} + \frac{1}{2}\mathbf{t}'\Sigma\mathbf{t})$.

Theorem 4.7

(a) $E\mathbf{X} = \mu$, $\operatorname{Var}\mathbf{X} = \Sigma$, and the distribution depends only on (μ, Σ). (b) If Σ is nonsingular,

$$f(\mathbf{x}) = \det(2\pi\Sigma)^{-1/2} \exp\{-\tfrac{1}{2}(\mathbf{x}-\mu)'\Sigma^{-1}(\mathbf{x}-\mu)\}.$$

(c) $B\mathbf{X} + \mathbf{c} \sim N(B\mu + \mathbf{c}, B\Sigma B')$; in particular every marginal and every linear combination is normal. (d) Components (or subvectors) are independent iff they are uncorrelated.

Proof. (a) $M(\mathbf{t}) = Ee^{\mathbf{t}'(A\mathbf{Z}+\mu)} = e^{\mathbf{t}'\mu} M_{\mathbf{Z}}(A'\mathbf{t}) = \exp(\mathbf{t}'\mu + \frac{1}{2}\mathbf{t}'AA'\mathbf{t})$, which involves only μ and Σ; the cgf gives the mean and covariance. (b) With $A = \Sigma^{1/2}$ (square, invertible), apply Theorem 4.1 to $\mathbf{x} = A\mathbf{z} + \mu$: $|J| = \det(A)^{-1}$ and $\mathbf{z}'\mathbf{z} = (\mathbf{x}-\mu)'\Sigma^{-1}(\mathbf{x}-\mu)$. (c) $B\mathbf{X} + \mathbf{c} = (BA)\mathbf{Z} + (B\mu + \mathbf{c})$. (d) If $\Sigma = \begin{pmatrix} \Sigma_{11} & 0 \\ 0 & \Sigma_{22} \end{pmatrix}$, the mgf factors into the mgfs of the two subvectors. ■

Independence-iff-uncorrelated is special to *jointly* normal vectors: if $Z \sim N(0,1)$ and $W = \pm Z$ with an independent fair sign, W is normal and uncorrelated with Z but not independent of it.

Theorem 4.8 (Conditional distributions)

Partition $\mathbf{X} = (\mathbf{X}_1, \mathbf{X}_2)$ with Σ_{22} nonsingular. Then

$$\mathbf{X}_1 \mid \mathbf{X}_2 = \mathbf{x}_2 \;\sim\; N\big(\mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(\mathbf{x}_2 - \mu_2),\; \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}\big).$$

Proof. Let $\mathbf{W} = \mathbf{X}_1 - \Sigma_{12}\Sigma_{22}^{-1}\mathbf{X}_2$. Then $\operatorname{Cov}(\mathbf{W}, \mathbf{X}_2) = \Sigma_{12} - \Sigma_{12} = 0$, so $\mathbf{W}$ is independent of $\mathbf{X}_2$ by Theorem 4.7(d). Its law is $N(\mu_1 - \Sigma_{12}\Sigma_{22}^{-1}\mu_2, \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})$. Write $\mathbf{X}_1 = \mathbf{W} + \Sigma_{12}\Sigma_{22}^{-1}\mathbf{X}_2$ and condition on $\mathbf{X}_2 = \mathbf{x}_2$. ■

For a bivariate normal: $E(Y \mid X = x) = \mu_2 + \rho\frac{\sigma_2}{\sigma_1}(x - \mu_1)$ is linear, and $\operatorname{Var}(Y \mid X) = \sigma_2^2(1-\rho^2)$ is constant.

Theorem 4.9 (Quadratic forms)

(a) If $\mathbf{X} \sim N_k(\mu, \Sigma)$ with Σ nonsingular, then $(\mathbf{X}-\mu)'\Sigma^{-1}(\mathbf{X}-\mu) \sim \chi^2(k)$. (b) If $\mathbf{Z} \sim N_n(\mathbf{0}, I)$ and A is symmetric idempotent ($A^2 = A$) of rank r, then $\mathbf{Z}'A\mathbf{Z} \sim \chi^2(r)$. (c) If moreover $AB = 0$ for symmetric idempotent B, then $\mathbf{Z}'A\mathbf{Z}$ and $\mathbf{Z}'B\mathbf{Z}$ are independent.

Proof. (a) $\mathbf{Z} = \Sigma^{-1/2}(\mathbf{X} - \mu) \sim N_k(\mathbf{0}, I)$ and the form equals $\mathbf{Z}'\mathbf{Z}$. (b) Diagonalize $A = P' \operatorname{diag}(I_r, 0) P$ with P orthogonal (the eigenvalues of an idempotent matrix are 0 or 1); $\mathbf{Y} = P\mathbf{Z} \sim N(\mathbf{0}, I)$ and $\mathbf{Z}'A\mathbf{Z} = Y_1^2 + \cdots + Y_r^2$. (c) $A\mathbf{Z}$ and $B\mathbf{Z}$ are jointly normal with $\operatorname{Cov} = AB' = AB = 0$, hence independent; and $\mathbf{Z}'A\mathbf{Z} = \|A\mathbf{Z}\|^2$, $\mathbf{Z}'B\mathbf{Z} = \|B\mathbf{Z}\|^2$. ■

Theorem 4.2 is the special case $A = I - \frac{1}{n}\mathbf{1}\mathbf{1}'$ (rank $n-1$), $B = \frac{1}{n}\mathbf{1}\mathbf{1}'$. The one-way-model statement after Corollary 4.4 follows the same way, with the projection onto group means minus the projection onto the grand mean (rank $k-1$).